

FUNDAMENTALS

1. What is coding and overview of Python Language

What is coding?

Coding is using a programming language to make a program, website, or game, etc., that helps simplify the way users interact with laptops or connect easily with other electronic devices.

Python Programming Language Overview for DE

Why Python is a perfect companion for Data Engineering (DE)? It has three main reasons: **Vast Ecosystem**, **Speed Of Coding** (due to its easily readable syntax), and being a **Glue Language**.

- **Vast Ecosystem:** Python supports users with a wide range of libraries which are perfect for DE, such as:
 - **Pandas** (Data Manipulation)
 - **PySpark** (Big Data Processing)
 - **Airflow** (Workflow Orchestration)
 - **Speed Of Coding:** Python syntax is easy to write and read.
 - *Example:* When you code an algorithm in C++ or Java, it may contain 50 lines of code, but in Python, you might only have to write 5 or 10 lines.
 - This allows you to build a pipeline faster and with less complexity.
 - **Glue Language:** It can connect disconnected systems easily (APIs, Databases, Cloud Services, etc.).
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2. Decomposition (Phân Chia Bài Toán)

Meaning: Decomposition is the process of breaking down a complex problem into smaller, sequential, and logical steps that a computer can understand and execute exactly.

Example: Collect Data base on (E-T-L)

Dividing to 3 independent parts:

1. **Extract (Trích xuất):** Only focus on collecting data.
2. **Transform (Chuyển đổi):** Only focus on cleaning and fining data.
3. **Load (Nạp):** Only focus on pushing data into storage(Data Warehouse).